

Interface Control Document

Vessel Autonomy Interfaces

Date	2026-02-10
Requesting System	Vessel Autonomy

Target System	Protocol	Data Format
Mission Autonomy	gRPC over HTTP/2	Protocol Buffers v3
Navigation Autonomy	UDP Multicast	NMEA 0183
Anduril Service Nav. Data	UDP Multicast	NMEA 0183
M&A Autonomy	TBD	TBD
D&S Autonomy	TBD	TBD

Surrogate ICD

Confidential

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Overview

NOTICE

This ICD is a **preliminary draft** prepared for initial coordination purposes. All interfaces defined herein, including message structures, payload fields, transport protocols, and data formats, are subject to change through ongoing discussions among stakeholders. No part of this document should be considered final or binding until formally agreed upon by all parties.

Interface to Mission Autonomy

This part defines all interfaces between Vessel Autonomy and Mission Autonomy, covering health monitoring, mode management, sensor configuration, failure response, damage reporting, waypoint commands, zone restrictions, and depth data.

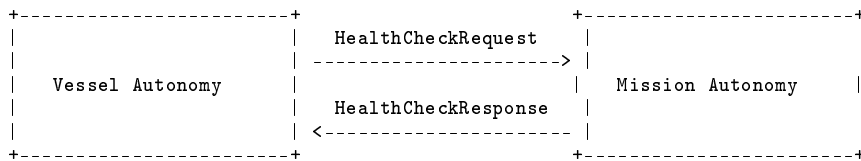
1 Health Check

1.1 Overview

The Vessel Autonomy system performs periodic health checks with the Mission Autonomy system to maintain real-time connection status. If a fault occurs in the target system or no response is received, an alert is immediately generated and delivered to the operator.

- Vessel Autonomy periodically sends health check requests to Mission Autonomy.
- Mission Autonomy returns its system status and subsystem statuses in the response.
- If no response is received or a fault condition is detected, Vessel Autonomy generates an alert.
- The generated alert is immediately delivered to the operator.

1.2 Interface Diagram



1.3 Service Definitions

1.3.1 HealthCheckService

A service for verifying the status of Mission Autonomy. Invoked from Vessel Autonomy toward Mission Autonomy.

RPC Name	Pattern	Request	Response	Description
CheckHealth	Unary	HealthCheckRequest	HealthCheckResponse	One-time status query
StreamHealth-Check	Bidir. Streaming	stream HealthCheckRequest	stream HealthCheckResponse	Periodic status check. Generates alert on connection loss or no response

1.3.2 AlertService

A service for delivering alerts to the operator when a fault is detected as a result of health checks.

RPC Name	Pattern	Request	Response	Description
SendAlert	Unary	AlertNotification	AlertAck	Send a single alert
Subscribe-Alerts	Server Streaming	AlertSubscribeRequest	stream AlertNotification	Operator console subscribes to real-time alert stream

1.4 Enumerations

1.4.1 SystemStatus

Represents the overall status of a system.

Value	Code	Description
SYSTEM_STATUS_UNSPECIFIED	0	Unknown (default)
SYSTEM_STATUS_HEALTHY	1	Operating normally
SYSTEM_STATUS_DEGRADED	2	Degraded performance
SYSTEM_STATUS_FAULT	3	Fault detected
SYSTEM_STATUS_OFFLINE	4	Offline / No response

1.4.2 ConnectionState

Represents the connection state between systems.

Value	Code	Description
CONNECTION_STATE_UNSPECIFIED	0	Unknown (default)
CONNECTION_STATE_CONNECTED	1	Connection active
CONNECTION_STATE_DISCONNECTED	2	Connection lost
CONNECTION_STATE_TIMEOUT	3	Response timeout

1.4.3 AlertSeverity

Represents the severity level of an alert.

Value	Code	Description
ALERT_SEVERITY_UNSPECIFIED	0	Unknown (default)
ALERT_SEVERITY_INFO	1	Informational
ALERT_SEVERITY_WARNING	2	Warning – attention required
ALERT_SEVERITY_CRITICAL	3	Critical – immediate action required

1.4.4 AlertType

Represents the type of alert raised.

Value	Code	Description
ALERT_TYPE_UNSPECIFIED	0	Unknown (default)
ALERT_TYPE_NO_RESPONSE	1	No response (timeout)
ALERT_TYPE_SYSTEM_FAULT	2	System fault detected
ALERT_TYPE_DEGRADED	3	Degraded performance detected
ALERT_TYPE_CONNECTION_LOST	4	Connection lost
ALERT_TYPE_RECOVERED	5	Fault recovered

1.5 Message Definitions

1.5.1 HealthCheckRequest

Request message sent from Vessel Autonomy to Mission Autonomy.

No.	Field	Type	Req.	Description
1	requester_id	string	Yes	Requesting system identifier (e.g. "vessel-autonomy-01")
2	request_timestamp	Timestamp	Yes	Request timestamp (UTC)
3	sequence_number	uint64	Yes	Sequence number; increments from 1 for periodic checks

1.5.2 HealthCheckResponse

Response message returned from Mission Autonomy to Vessel Autonomy.

No.	Field	Type	Req.	Description
1	responder_id	string	Yes	Responding system identifier (e.g. "mission-autonomy-01")
2	response_timestamp	Timestamp	Yes	Response timestamp (UTC)
3	sequence_number	uint64	Yes	Sequence number matching the request
4	system_status	SystemStatus	Yes	Overall system status
5	connection_state	ConnectionState	Yes	Current connection state
6	subsystem_statuses	SubsystemStatus[]	No	Detailed subsystem statuses
7	status_message	string	No	Human-readable status description

1.5.3 SubsystemStatus

Represents the status of an individual subsystem within Mission Autonomy.

No.	Field	Type	Req.	Description
1	subsystem_name	string	Yes	Subsystem name (e.g. "navigation", "sensor")
2	status	SystemStatus	Yes	Status of the subsystem
3	detail	string	No	Detailed description

1.5.4 AlertNotification

Alert message delivered to the operator when a health check failure or fault is detected.

No.	Field	Type	Req.	Description
1	alert_id	string	Yes	Unique alert ID (UUID recommended)
2	timestamp	Timestamp	Yes	Alert timestamp (UTC)
3	severity	AlertSeverity	Yes	Alert severity level
4	target_system_id	string	Yes	Faulted target system ID
5	alert_type	AlertType	Yes	Alert type code
6	title	string	Yes	Alert title
7	description	string	No	Detailed description
8	related_sequence_number	uint64	No	Related health check sequence no.
9	detected_status	SystemStatus	Yes	Detected system status
10	detected_connection_state	ConnectionState	Yes	Detected connection state

1.5.5 AlertAck

Acknowledgement response confirming that the operator has received the alert.

No.	Field	Type	Req.	Description
1	alert_id	string	Yes	Alert ID being acknowledged
2	ack_timestamp	Timestamp	Yes	Acknowledgement timestamp (UTC)
3	acknowledged	bool	Yes	Whether acknowledged (true = confirmed)

1.5.6 AlertSubscribeRequest

Request message used by an operator console to subscribe to a real-time alert stream.

No.	Field	Type	Req.	Description
1	subscriber_id	string	Yes	Subscriber ID (e.g. operator console ID)
2	min_severity	AlertSeverity	No	Minimum severity filter
3	target_system_ids	string[]	No	Filter by target systems; empty = all

2 Master Mode Change

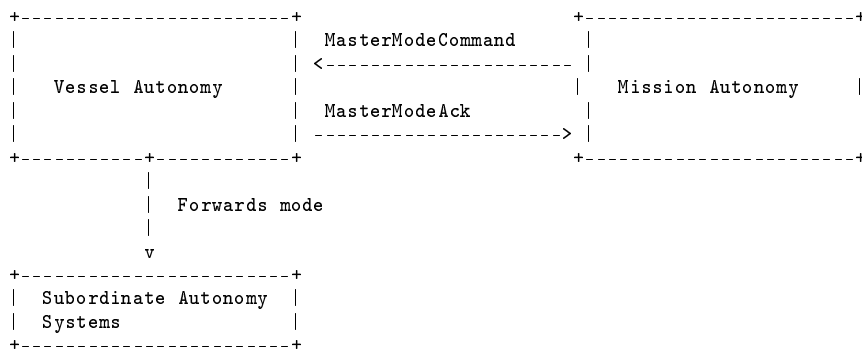
2.1 Overview

The vessel's top-level operating mode is set to either **Manual** or **Autonomous**. Mission Autonomy determines the master mode and pushes the mode change command to Vessel Autonomy. Once confirmed, Vessel Autonomy receives the command and forwards it to the subordinate autonomy systems.

During the **Surrogate Demo** phase, Vessel Autonomy (KSOE) only receives updates related to changes in the master mode; it does not initiate mode changes.

- Mission Autonomy sends a master mode change notification to Vessel Autonomy.
- Vessel Autonomy receives and applies the new mode (Manual or Autonomous).
- Vessel Autonomy forwards the mode to all subordinate systems.
- Vessel Autonomy acknowledges the mode change back to Mission Autonomy.

2.2 Interface Diagram



2.3 Operating Modes (Table 3-1)

Detailed operating modes of Vessel Autonomy and subordinate systems.

Operating Mode	Control Authority	Key Characteristics
Manual Mode	Onboard operator direct control	• Navigation Autonomy disabled • No remote control • (TBD: Main & Auxiliary, Damage & Safety characteristics)
Autonomous Mode	Lattice (C2) control (Offboard)	• Navigation Autonomy on standby or executing a mission • Collision avoidance active only during “track control” • Under review: enabling collision avoidance in Auto Heading mode • Plan to maintain collision avoidance during post-mission standby (“donuts”) • (TBD: Main & Auxiliary, Damage & Safety characteristics)

2.4 Service Definition

2.4.1 MasterModeService

A service through which Mission Autonomy pushes master mode changes to Vessel Autonomy. Vessel Autonomy acts as a **receiver only** in this interface.

RPC Name	Pattern	Request	Response	Description
ChangeMasterMode	Unary	MasterModeCommand	MasterModeAck	Mission Autonomy sends a mode change command. Vessel Autonomy applies the mode and returns acknowledgement
SubscribeMasterMode	Server Streaming	MasterModeSubscribeRequest	stream MasterModeNotification	Vessel Autonomy subscribes to a stream of master mode change events from Mission Autonomy

2.5 Enumerations

2.5.1 MasterMode

Represents the vessel's top-level operating mode.

Value	Code	Description
MASTER_MODE_UNSPECIFIED	0	Unknown (default)
MASTER_MODE_MANUAL	1	Manual Mode – onboard operator direct control
MASTER_MODE_AUTONOMOUS	2	Autonomous Mode – Lattice (C2) offboard control

2.5.2 ModeChangeResult

Represents the result of a mode change operation.

Value	Code	Description
MODE_CHANGE_RESULT_UNSPECIFIED	0	Unknown (default)
MODE_CHANGE_RESULT_SUCCESS	1	Mode change applied successfully
MODE_CHANGE_RESULT_REJECTED	2	Mode change rejected (e.g. safety constraint)
MODE_CHANGE_RESULT_FAILED	3	Mode change failed (internal error)

2.6 Message Definitions

2.6.1 MasterModeCommand

Command message sent **from Mission Autonomy to Vessel Autonomy** to change the vessel's master operating mode.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Unique command identifier (UUID recommended)
2	timestamp	Timestamp	Yes	Command timestamp (UTC)
3	requested_mode	MasterMode	Yes	The target mode to switch to (MANUAL or AUTONOMOUS)
4	issued_by	string	Yes	Identifier of the issuing system (e.g. "mission-autonomy-01")

2.6.2 MasterModeAck

Acknowledgement message returned **from Vessel Autonomy to Mission Autonomy** after processing a mode change command.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Matching command ID from the request
2	timestamp	Timestamp	Yes	Acknowledgement timestamp (UTC)
3	result	ModeChangeResult	Yes	Result of the mode change operation
4	current_mode	MasterMode	Yes	The mode now active on the vessel
5	message	string	No	Human-readable result message (e.g. rejection reason)

2.6.3 MasterModeSubscribeRequest

Request message used by Vessel Autonomy to subscribe to master mode change events.

No.	Field	Type	Req.	Description
1	subscriber_id	string	Yes	Subscriber identifier (e.g. "vessel-autonomy-01")

2.6.4 MasterModeNotification

Streaming notification sent from Mission Autonomy whenever the master mode changes.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Unique identifier for this mode change event
2	timestamp	Timestamp	Yes	Event timestamp (UTC)
3	new_mode	MasterMode	Yes	The new master mode (MANUAL or AUTONOMOUS)
4	previous_mode	MasterMode	Yes	The previous master mode
5	issued_by	string	Yes	Identifier of the issuing system

3 Radar/PTZ Mode Select (During PSA Test)

3.1 Overview

During the PSA test, the radar and PTZ camera are each operated in two modes (**Auto** and **Manual**). The operating mode is configured by Mission Autonomy, and when a mode change occurs, the updated mode must be received by Vessel Autonomy in real time.

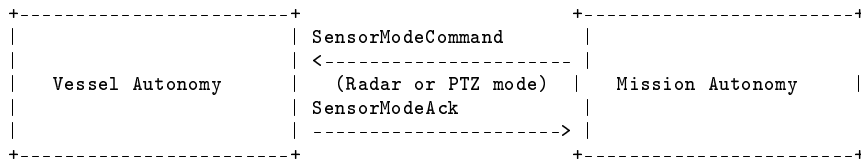
Vessel Autonomy acts as a **receiver only**; it does not initiate mode changes for radar or PTZ.

- Mission Autonomy determines and pushes the radar operating mode (Auto/Manual) to Vessel Autonomy.
- Mission Autonomy determines and pushes the PTZ operating mode (Auto/Manual) to Vessel Autonomy.
- Each device mode is managed independently — radar and PTZ may be in different modes simultaneously.
- Vessel Autonomy acknowledges each mode change back to Mission Autonomy.

Mode characteristics summary:

Device	Mode	Description
Radar	Auto	Radar performs automatic target tracking. Vessel Autonomy receives TTM (Tracked Target Message) data for targets designated by Mission Autonomy.
Radar	Manual	Radar outputs raw data (Spoke and Blob). Vessel Autonomy receives raw data and selectively processes targets designated by Mission Autonomy.
PTZ	Auto	PTZ camera automatically captures surrounding targets periodically. Vessel Autonomy receives still images or video of detected targets. (TBD: via HiNAS or direct connection)
PTZ	Manual	Mission Autonomy manually controls pan/tilt/zoom to track a target. Vessel Autonomy receives still images or video of the designated target. (TBD: via HiNAS or direct connection)

3.2 Interface Diagram



3.3 Service Definition

3.3.1 SensorModeService

A service through which Mission Autonomy pushes radar and PTZ mode changes to Vessel Autonomy. Vessel Autonomy acts as a **receiver only**.

RPC Name	Pattern	Request	Response	Description
ChangeSensorMode	Unary	SensorModeCommand	SensorModeAck	Mission Autonomy sends a sensor mode change command. Vessel Autonomy applies and acknowledges
SubscribeSensorMode	Server Streaming	SensorModeSubscribeRequest	stream SensorModeNotification	Vessel Autonomy subscribes to real-time sensor mode change events

3.4 Enumerations

3.4.1 DeviceType

Identifies the sensor device whose mode is being changed.

Value	Code	Description
DEVICE_TYPE_UNSPECIFIED	0	Unknown (default)
DEVICE_TYPE_RADAR	1	Radar
DEVICE_TYPE_PTZ	2	PTZ Camera

3.4.2 SensorOperatingMode

Represents the operating mode of a sensor device.

Value	Code	Description
SENSOR_MODE_UNSPECIFIED	0	Unknown (default)
SENSOR_MODE_AUTO	1	Auto mode
SENSOR_MODE_MANUAL	2	Manual mode

3.4.3 SensorModeChangeResult

Represents the result of a sensor mode change operation.

Value	Code	Description
SENSOR_MODE_CHANGE_UNSPECIFIED	0	Unknown (default)
SENSOR_MODE_CHANGE_SUCCESS	1	Mode change applied successfully
SENSOR_MODE_CHANGE_REJECTED	2	Mode change rejected
SENSOR_MODE_CHANGE_FAILED	3	Mode change failed (internal error)

3.5 Message Definitions

3.5.1 SensorModeCommand

Command message sent **from Mission Autonomy to Vessel Autonomy** to change the operating mode of a radar or PTZ device.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Unique command identifier (UUID recommended)
2	timestamp	Timestamp	Yes	Command timestamp (UTC)
3	device_type	DeviceType	Yes	Target device (RADAR or PTZ)
4	requested_mode	SensorOperatingMode	Yes	Target mode (AUTO or MANUAL)
5	issued_by	string	Yes	Identifier of the issuing system

3.5.2 SensorModeAck

Acknowledgement returned **from Vessel Autonomy to Mission Autonomy** after processing a sensor mode change command.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Matching command ID from the request
2	timestamp	Timestamp	Yes	Acknowledgement timestamp (UTC)
3	device_type	DeviceType	Yes	Device that was changed
4	result	SensorModeChangeResult	Yes	Result of the mode change
5	current_mode	SensorOperatingMode	Yes	Mode now active on the device
6	message	string	No	Human-readable result message

3.5.3 SensorModeSubscribeRequest

Request message used by Vessel Autonomy to subscribe to sensor mode change events.

No.	Field	Type	Req.	Description
1	subscriber_id	string	Yes	Subscriber identifier (e.g. "vessel-autonomy-01")
2	device_filter	DeviceType[]	No	Filter by device type; empty = all devices

3.5.4 SensorModeNotification

Streaming notification sent from Mission Autonomy whenever a sensor mode changes.

No.	Field	Type	Req.	Description
1	command_id	string	Yes	Unique identifier for this event
2	timestamp	Timestamp	Yes	Event timestamp (UTC)
3	device_type	DeviceType	Yes	Device whose mode changed
4	new_mode	SensorOperatingMode	Yes	The new mode (AUTO or MANUAL)
5	previous_mode	SensorOperatingMode	Yes	The previous mode
6	issued_by	string	Yes	Identifier of the issuing system

4 Diagnostic Data Transmission

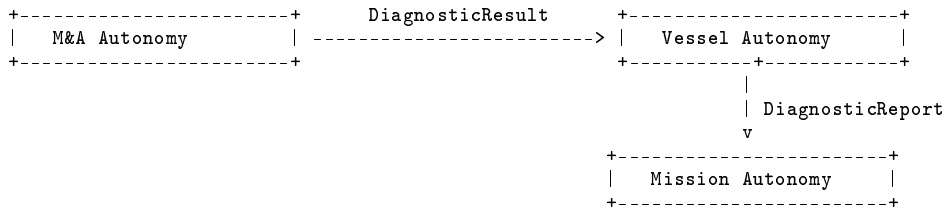
4.1 Overview

Vessel Autonomy receives diagnostic results from Main & Auxiliary (M&A) Autonomy, including equipment fault diagnosis status. This upstream flow is one-way: M&A Autonomy transmits diagnostic data to Vessel Autonomy.

Vessel Autonomy then consolidates and forwards the diagnostic package to Mission Autonomy through the external interface defined in this ICD.

- M&A Autonomy generates per-equipment diagnostic results.
- M&A Autonomy reports diagnostic results to Vessel Autonomy.
- Vessel Autonomy relays diagnostic results to Mission Autonomy.
- Mission Autonomy receives periodic updates for operational awareness.

4.2 Interface Diagram



4.3 Service Definition

4.3.1 DiagnosticDataService

Service for forwarding equipment diagnostic results from Vessel Autonomy to Mission Autonomy.

RPC Name	Pattern	Request	Response	Description
ReportDiagnosticData	Unary	DiagnosticReport	DiagnosticAck	Vessel Autonomy forwards periodic diagnostic results to Mission Autonomy
StreamDiagnosticData	Server Streaming	DiagnosticSubscribeRequest	stream DiagnosticReport	Mission Autonomy subscribes to periodic diagnostic updates

4.4 Diagnostic Result Data Transmission (Reference: 3.4.2.2)

Independent diagnostic results are generated for each piece of equipment, providing a comprehensive status overview. Each diagnostic package includes:

- Equipment identification information
- Current operational status (ON / OFF)
- Representative diagnostic score
- Four-level diagnostic rating (NORMAL / CAUTION / WARNING / DANGER)
- Detailed subsystem-level diagnostic scores (provided according to the scope of available VMS data)

Diagnostic results are periodically updated and delivered to Mission Autonomy through Vessel Autonomy. For the four main engines, the update interval is approximately 10 seconds (Large Surrogate platform only).

Table 3-3: Diagnostic Score Ranges and Status Levels

Score Range	Status Level	Description
80–100	Normal	• Equipment is operating normally
60–80	Caution	• Deviates from normal range • Full load or harsh operation should be limited
40–60	Warning	• Clear abnormal condition detected • Load reduction is strongly recommended
Below 40	Danger	• Equipment stop is recommended • Switch to redundancy if possible • RTB (Return to Base) is recommended if mission capability is impacted

4.5 Message Definitions

4.5.1 EquipmentDiagnostic

Represents a diagnostic result for a single equipment item.

No.	Field	Type	Req.	Description
1	equipment_id	string	Yes	Unique equipment identifier
2	equipment_name	string	No	Human-readable equipment name
3	operational_status	string	Yes	Current state ("ON" / "OFF")
4	diagnostic_score	double	Yes	Representative score (0-100)
5	diagnostic_level	string	Yes	Four-level rating (Normal/Caution/Warning/Danger)
6	subsystem_scores	string	No	Encoded subsystem-level score set (VMS scope dependent)
7	updated_at	Timestamp	Yes	Diagnostic update timestamp (UTC)

4.5.2 DiagnosticReport

Diagnostic report forwarded from Vessel Autonomy to Mission Autonomy.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Unique report ID
2	timestamp	Timestamp	Yes	Report timestamp (UTC)
3	source_system_id	string	Yes	Usually Vessel Autonomy system ID
4	target_system_id	string	Yes	Mission Autonomy system ID
5	update_interval_sec	uint32	No	Update interval in seconds (e.g. 10)
6	equipment_results	EquipmentDiagnostic[]	Yes	Per-equipment diagnostic results

4.5.3 DiagnosticAck

Acknowledgement returned from Mission Autonomy to Vessel Autonomy.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Matching report ID
2	timestamp	Timestamp	Yes	Acknowledgement timestamp (UTC)
3	accepted	bool	Yes	Whether the report was accepted
4	message	string	No	Optional status message

4.5.4 DiagnosticSubscribeRequest

Subscription request for periodic diagnostic reports.

No.	Field	Type	Req.	Description
1	subscriber_id	string	Yes	Subscriber identifier
2	requested_period_sec	uint32	No	Desired update interval in seconds
3	equipment_filter	string[]	No	Optional equipment ID filter

5 Operational Recommendation Response

5.1 Overview

When a subordinate system (e.g. Main & Auxiliary Autonomy) detects a failure, it reports the failure to Vessel Autonomy. Vessel Autonomy then consolidates the failure report and forwards it to Mission Autonomy.

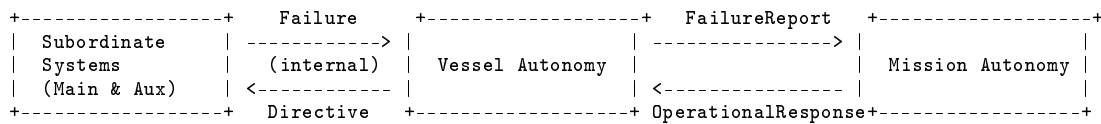
Mission Autonomy evaluates the severity of the failure and selects an appropriate response strategy. The M&A failure scenarios are categorized into two levels:

- **Scenario 1 – Engine Degradation:** Engine performance degrades over time. Mission Autonomy commands a **Ship Speed Reduction** to reduce load on the affected engine(s).
- **Scenario 2 – Critical Failure & Emergency Response:** A critical or catastrophic failure is detected. Mission Autonomy selects one of the emergency response options (Option B or C).

The selected response is sent back to Vessel Autonomy, which applies the directive and forwards it to the subordinate systems.

- Subordinate systems report failures to Vessel Autonomy.
- Vessel Autonomy sends a **FailureReport** to Mission Autonomy.
- Mission Autonomy evaluates severity and selects an operational response option.
- Mission Autonomy sends **OperationalResponse** back to Vessel Autonomy.
- Vessel Autonomy acknowledges and forwards the directive to subordinate systems.

5.2 Interface Diagram



5.3 Operational Response Options (Table 3-2)

Scenario 1 – Engine Degradation Response:

Option	Name	Description
Speed Reduction	Ship Speed Reduction	• Reduce vessel speed to lower engine load • Applicable when engine performance degrades over time • Mission Autonomy commands target speed reduction • Navigation Autonomy adjusts route/speed accordingly

Scenario 2 – Critical Failure & Emergency Response:

Option	Name	Description
Option A	(TBD – Idle the Faulty Engine Only)	• TBD – Low feasibility; risk of asymmetric thrust • Not recommended under current design constraints
Option B	Ship Stop & All Engines Idle (Manual Mode)	• Switch the vessel to Manual Mode (HiNAS does not support Ship Stop) • Stop the vessel and set all engines to Idle • Most conservative and safest response
Option C	Low-Power Return to Base (RTB)	• Operate all engines at minimum safe output • Mission Autonomy reassigns the mission as RTB • Executable without switching to Manual Mode • HiNAS performs the mission under limited propulsion

5.4 Service Definition

5.4.1 OperationalResponseService

A service through which Vessel Autonomy reports failures to Mission Autonomy and receives the selected operational response.

RPC Name	Pattern	Request	Response	Description
ReportFailure	Unary	FailureReport	OperationalResponse	Vessel Autonomy reports a failure to Mission Autonomy. Mission Autonomy evaluates and returns the selected response option
Acknowledge-Response	Unary	ResponseAck	ResponseAckConfirm	Vessel Autonomy confirms that the operational response has been applied

5.5 Enumerations

5.5.1 FailureSeverity

Represents the severity of a failure reported by a subordinate system.

Value	Code	Description
FAILURE_SEVERITY_UNSPECIFIED	0	Unknown (default)
FAILURE_SEVERITY_LOW	1	Low – minor degradation
FAILURE_SEVERITY_MEDIUM	2	Medium – noticeable impact on operations
FAILURE_SEVERITY_HIGH	3	High – significant risk to vessel
FAILURE_SEVERITY_CRITICAL	4	Critical – immediate action required

5.5.2 FailureSource

Identifies the specific equipment that originated the failure. Due to Surrogate data limitations, the diagnostic target is currently limited to Main Engines (ME 1–4).

Value	Code	Description
FAILURE_SOURCE_UNSPECIFIED	0	Unknown (default)
FAILURE_SOURCE_MAIN_ENGINE_1	1	Main Engine #1
FAILURE_SOURCE_MAIN_ENGINE_2	2	Main Engine #2
FAILURE_SOURCE_MAIN_ENGINE_3	3	Main Engine #3
FAILURE_SOURCE_MAIN_ENGINE_4	4	Main Engine #4
FAILURE_SOURCE_AUX_ENGINE	5	Auxiliary engine system (TBD)
FAILURE_SOURCE_OTHER	6	Other subordinate system

5.5.3 ResponseOption

Represents the operational response option selected by Mission Autonomy.

Value	Code	Description
RESPONSE_OPTION_UNSPECIFIED	0	Unknown (default)
RESPONSE_OPTION_SPEED_REDUCTION	1	Scenario 1: Ship Speed Reduction (engine degradation)
RESPONSE_OPTION_A	2	Scenario 2: TBD – Idle faulty engine only (low feasibility)
RESPONSE_OPTION_B	3	Scenario 2: Ship Stop & All Engines Idle (Manual Mode)
RESPONSE_OPTION_C	4	Scenario 2: Low-Power Return to Base (RTB)

5.6 Message Definitions

5.6.1 FailureReport

Failure report sent from Vessel Autonomy to Mission Autonomy. Vessel Autonomy consolidates failure information received from subordinate systems and forwards it to Mission Autonomy for evaluation.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Unique report identifier (UUID recommended)
2	timestamp	Timestamp	Yes	Report timestamp (UTC)
3	reporter_id	string	Yes	Reporting system identifier (e.g. "vessel-autonomy-01")
4	failure_source	FailureSource	Yes	Subordinate system that detected the failure
5	severity	FailureSeverity	Yes	Failure severity level
6	failure_code	string	Yes	Machine-readable failure code
7	description	string	No	Human-readable failure description
8	affected_engines	string[]	No	List of affected engine identifiers

5.6.2 OperationalResponse

Response returned from Mission Autonomy to Vessel Autonomy containing the selected operational response strategy.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Matching report ID from the failure report
2	timestamp	Timestamp	Yes	Response timestamp (UTC)
3	selected_option	ResponseOption	Yes	Selected response option (Speed Reduction, A, B, or C)
4	issued_by	string	Yes	Identifier of the issuing system
5	rationale	string	No	Explanation for the selected option
6	requires_mode_change	bool	Yes	Whether the response requires a master mode change (e.g. Option B → Manual)
7	target_mode	string	No	Target master mode if mode change is required (e.g. "MANUAL")

5.6.3 ResponseAck

Acknowledgement sent **from Vessel Autonomy to Mission Autonomy** confirming that the operational response has been received and applied.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Matching report ID
2	timestamp	Timestamp	Yes	Acknowledgement timestamp (UTC)
3	applied	bool	Yes	Whether the response was successfully applied
4	message	string	No	Human-readable status message

5.6.4 ResponseAckConfirm

Confirmation returned **from Mission Autonomy to Vessel Autonomy** that the acknowledgement was received.

No.	Field	Type	Req.	Description
1	report_id	string	Yes	Matching report ID
2	timestamp	Timestamp	Yes	Confirmation timestamp (UTC)
3	confirmed	bool	Yes	Whether the acknowledgement was accepted

6 Damage & Safety Event Report

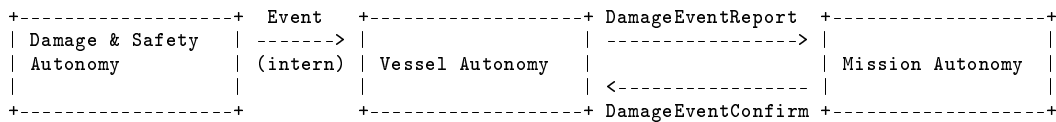
6.1 Overview

The Damage & Safety Autonomy system monitors the vessel for onboard damage and safety-related abnormalities such as fire and smoke. When an event is detected, it reports the information to Vessel Autonomy.

Vessel Autonomy consolidates the event data and forwards it to Mission Autonomy so that Mission Autonomy can verify the situation. Mission Autonomy returns a confirmation acknowledging that the event has been received and noted.

- Damage & Safety Autonomy detects fire or smoke and reports the event to Vessel Autonomy.
- Vessel Autonomy forwards the event as a **DamageEventReport** to Mission Autonomy.
- Mission Autonomy verifies the report and returns a **DamageEventConfirm** (acknowledgement only).
- Mission Autonomy does **not** send a corrective action in this interface; it only confirms receipt.

6.2 Interface Diagram



6.3 Detectable Damage Types and Reported Information (Table 3-5)

Category	Items
Detectable Damage Types	• Fire • Smoke
Information Provided	• Location of occurrence • Time of occurrence • Time of resolution • Detected object information (type, coordinates, confidence) • Anomaly map (highlighting abnormal regions) • Quantified damage ratio (damaged pixels / total pixels)

6.4 Service Definition

6.4.1 DamageEventService

A service through which Vessel Autonomy reports damage and safety events to Mission Autonomy. Mission Autonomy confirms receipt of the report.

RPC Name	Pattern	Request	Response	Description
ReportDamage-Event	Unary	DamageEventReport	DamageEventConfirm	Vessel Autonomy sends a damage/safety event. Mission Autonomy confirms receipt
StreamDamage-Events	Server Streaming	DamageEventSubscribeRequest	stream DamageEvent-Report	Mission Autonomy subscribes to a real-time stream of damage events from Vessel Autonomy

6.5 Enumerations

6.5.1 DamageType

Type of damage or safety abnormality detected.

Value	Code	Description
DAMAGE_TYPE_UNSPECIFIED	0	Unknown (default)
DAMAGE_TYPE_FIRE	1	Fire detected
DAMAGE_TYPE_SMOKE	2	Smoke detected

6.5.2 DamageEventStatus

Lifecycle status of a damage event.

Value	Code	Description
DAMAGE_EVENT_STATUS_UNSPECIFIED	0	Unknown (default)
DAMAGE_EVENT_STATUS_ACTIVE	1	Event is ongoing
DAMAGE_EVENT_STATUS_RESOLVED	2	Event has been resolved